Voronoi Homogeneity Analysis of Categorical Data

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TBD

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1 Introduction

1.1 Homogeneity Analysis

Homogeneity Analysis, a.k.a. *Multiple Correspondence Analysis*, can be introduced in many different ways. We refer to Guttman (1941), Tenenhaus and Young (1985), Bekker and De Leeuw (1988), Gifi (1990), Greenacre and Blasius (2006), Le Roux and Rouanet (2010), and De Leeuw (2023) for a bouquet of discussions and interpretations of the equations and algorithms defining the technique.

The data are measurements of n objects on m categorical variables. Variable j has k_j *categories*, which are the different unique values the variable assumes for the n objects. We also say that a category has a number of objects that "belong to" the category.

Here is a small example with $n = 10$ and $m = 3$, taken from Gifi (1990), page 65.

```
      [,1] [,2] [,3]
 [1,] "a"  "p"  "u"
 [2,] "b"  "q"  "v"
 [3,] "a"  "r"  "v"
 [4,] "a"  "p"  "u"
 [5,] "b"  "p"  "v"
 [6,] "c"  "p"  "v"
 [7,] "a"  "p"  "u"
 [8,] "a"  "p"  "v"
 [9,] "c"  "p"  "v"
[10,] "a"  "p"  "v"
```

Suppose the objects are mapped into n points of Euclidean space $\mathbb{R}^p$. For each variable we can construct an additional k_j points in the same space by computing for each category the centroids of all object-points that belong to it. Thus each variable defines its own plot. We then draw lines from the n object-points to the category-point they belong to and we obtain k_j "stars". Homogeneity Analysis maps the objects into space in such a way that the stars are as small as possible, where the size of the star is measured as the sum of squares of the distances between each category-points and the object points of the oibjects that belong to the category (summed over categories and over variables). In order to avoid the trivial solution in which all objects and categories are mapped into the same point Homogeneity Analysis requires that the *configuration* of object-points (an $n \times p$ matrix) is column-centered and orthonormal.

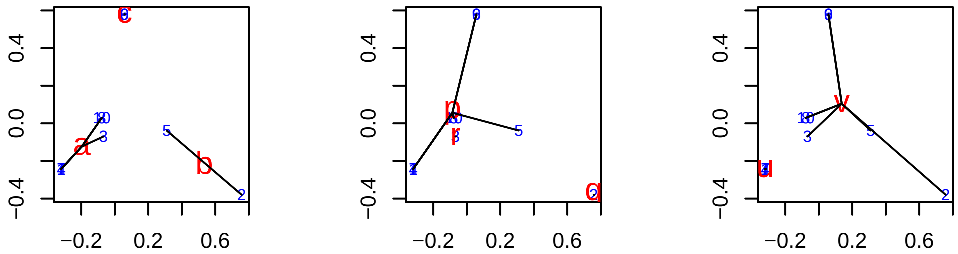
There are a huge number of variations (De Leeuw (2004)) possible on the same theme. Instead of the centroids we could use the Weber points to define our stars. Instead of Euclidean distance we could use any other distance on $\mathbb{R}^p$. Instead of the stars we could use the convex hulls or the minimum spanning trees, or any other way to visualize and quantify the compactness of the point clouds of each category. Any one of these choices defines a technique that could be called Homogeneity Analysis, but in this paper we mean the classical choice that uses stars, squared Euclidean distance, and orthonormal object-points.

The key mathematical object in Homogeneity Analysis is the *indicator matrix*, which is a classical way to code categorical data[1]. For a variable j with k_j categories it is an $n \times k_j$ binary matrix G_j in which each row i contains a single element equal to one, indicating into which category the object i maps for variable j. The cross product $D_j := G_j'G_j$ is diagonal, with the *marginal frequencies* of the categories on the diagonal. For two different variables the crossproduct $C_{j\ell} = G_j'G_\ell$ is the $k_j \times k_\ell$ *cross table* or *contingency table* of the two variables. Here are the three indicator matrices of our small example, next to each other.

```
        a b c    p q r    u v
 [1,] || 1 0 0 || 1 0 0 || 1 0 ||
 [2,] || 0 1 0 || 0 1 0 || 0 1 ||
 [3,] || 1 0 0 || 0 0 1 || 0 1 ||
 [4,] || 1 0 0 || 1 0 0 || 1 0 ||
 [5,] || 0 1 0 || 1 0 0 || 0 1 ||
 [6,] || 0 0 1 || 1 0 0 || 0 1 ||
 [7,] || 1 0 0 || 1 0 0 || 1 0 ||
 [8,] || 1 0 0 || 1 0 0 || 0 1 ||
 [9,] || 0 0 1 || 1 0 0 || 0 1 ||
[10,] || 1 0 0 || 1 0 0 || 0 1 ||
```

A Homogeneity Analysis in two dimensions produces the following three star plots. Objects are numbered and in BLUE, categories have bigger labels and are in RED. Note that there are ten objects, but the sets of objects $(1, 4, 7)$, $(8, 10)$, and $(6, 9)$ have the same profile and consequently get mapped to the same object-point. Thus there are at most six different object-points. The analysis minimizes the sum of squares of the length of all black lines in the three plots, under the condition that the matrix of object scores is column-centered and orthonormal. We can think of the results as a form of *graph drawing*, where the graph is the union of the m bipartite graphs of all m "belonging to" relations (Michailidis and De Leeuw (2001)).

[1]Since the indicator matrix is the core of the Gifi system we avoid the irreverent alternative "dummy coding".



We now deviate from the usual Gifi and French approaches (Husson et al. (2016)) and define Homogeneity Analysis as a form of non-metric multidimensional scaling, using the *stress* loss function (Kruskal (1964a), Kruskal (1964b)). Define[2]

$$\sigma(X,Y_1,\cdots,Y_m,\Delta_1,\cdots,\Delta_m)) := \sum_{j=1}^m\sum_{i=1}^n\sum_{\ell=1}^{k_j}(\delta_{i\ell}^j - d(x_i,y_\ell^j))^2, \tag{1}$$

with $d(.,.)$ Euclidean distance. Homogeneity analysis minimizes this loss over all configurations X with $X'X = I$, over all Y_j, and over all Δ_j that satisfy $\delta_{i\ell}^j = 0$ whenever $g_{i\ell}^j = 1$. All other elements of the Δ_j are only constrained to be non-negative.

If

$$\sigma_\star(X,Y_1,\cdots,Y_m) := \min_{\Delta_1,\cdots,\Delta_m}\sigma(X,Y_1,\cdots,Y_m,\Delta_1,\cdots,\Delta_m), \tag{2}$$

then

$$\sigma_\star(X,Y_1,\cdots,Y_m) = \sum_{j=1}^m\sum_{i=1}^n\sum_{\ell=1}^{k_j} g_{i\ell}^j d^2(x_i,y_\ell^j). \tag{3}$$

Next define

$$\sigma_{\star\star}(X) := \min_{Y_1,\cdots,Y_m}\sigma_\star(X;Y_1,\cdots,Y_m). \tag{4}$$

The minimum of (3) over the Y_j is attained at the centroids $\hat{Y}_j := D_j^{-1}G_j'X$, and if we substitute these optimal $\hat{Y}_j$ in (3) we see that $\sigma_{\star\star}(X)$ is indeed equal to the sum of squares of all lines defining the stars. Minimizing $\sigma_{\star\star}(X)$ over $X'X = I$, and thus minimizing (1) over (X, Y_j, Δ_j), defines Homogeneity Analysis.

Actually computing the solution turns out to be a straightforward eigenvalue-eigenvector problem. The details are in the publications we referenced at the beginning of this section.

[2] The symbol $:=$ is used for definitions

1.2 Voronoi Analysis

Although using loss function (1) makes Homogeneity Analysis a form of non-metric multi-dimensional scaling there are some important differences with more familiar applications of minimizing stress. There are no local minima in Homogeneity Analysis, because it tuirns out to be a symmetric eigenvalue problem. Also, and perhaps most importantly, we do not really expect the minimum loss function value to be zero, or even close to zero. The loss functions measures deviations from a very unrealistic model. Zero loss would imply all object-points coincide with all category-points they belong to. This can only be realized, except in trivial cases, if all object and category-points map into a single point, which is prohibited by the orthonormality constraint on the object scores. Thus we get a perfectly determinate and stable solution, at low computational cost, but by imposing unrealistic constraints and a rather arbitrary normalization. We can get around this seemingly unfortunate situation by admitting we are not fitting or testing a "model", but we are making drawings of the bipartite graphs that constitute the data.

The lack of a realistic model makes it interesting to generalize loss function (1) to

$$\sigma(X, Y_1, \cdots, Y_m, \Delta_1, ..., \Delta_m) := \sum_{j=1}^{m}\sum_{i=1}^{n}\sum_{\ell=1}^{k_j}(\delta_{i\ell}^j - d(x_i, y_\ell^j))^2, \tag{5}$$

with the constraints that if $g_{i\ell}^j = 1$ then merely $\delta_{i\ell}^j \leq \delta_{i\nu}^j$ for all $\nu = 1, \cdots, k_j$. This is significantly weaker than the homogeneity analysis constraints, which require the mn disparities $\delta_{i\ell}^j$ to be zero. If object i belongs to category ℓ of variable j we do no longer require the object-point to coincide with the category-point, but we merely require that the object-point is at least as close to category-point ℓ as to any of the other $k_j - 1$ category-points of that variable.

As a consequence, the plots corresponding with loss function (5) are also different. As in Homogeneity Analysis there is a plot for each variable j with both n object-points and k_j category-points. The k_j category-points define k_j *voronoi regions* that partition the space. Voronoi region $\mathcal{V}_\ell^j$ is the region of space where the points are at least as close to category-point y_ℓ^j as to any of the other category-points of variable j. And loss (4) is zero (for a variable) if all object-points are in the "correct" Voronoi region, i.e. in the Voronoi region of the category point corresponding with the category they belong to.

One (very inefficient, but conceptually easy) way of constructing Voronoi regions in the plane is to draw the perpendicular bisectors of the (hypothetical) lines connecting all pairs of category-points of a variable. In higher dimensions $p > 2$ the bisectors are planes of dimension $p - 1$. The convex polyhedral regions thus formed, some bounded and some unbounded, define the Voronoi regions. The literature on properties and computation on Voronoi regions is large, and we refer to the textbooks of Okabe et al. (2000) and Aurenhammer et al. (2013).

Since the category-point itself is obviously in its own Voronoi region and since Voronoi regions are convex it follows that if we make stars by connecting category-points with object points as in Homogeneity Analysis then we actually require each star to be in its Voronoi region. It is no longer true that a category-point is the centroid of points of the objects belonging to the category. The category-point can even be outside the convex hull of "its" object-points.

Voronoi Homogeneity Analysis of categorical data, as defined here, minimizes loss function (5) over X, over the Y_j and over the Δ_j satisfying the ordinal constraints. It is consequently another form of non-metric (ordinal) multidimensional scaling. To be more precise, we have m linked non-metric multidimensional unfolding problems, which are linked because they share the same X. The data are binary, and we use Kruskal's primary approach to ties (De Leeuw (1977b)).

The constraint $X'X = I$ from Homogeneity Analysis is no longer needed, but some form of normalization is still necessary. We still need to exclude the trivial solution in which all points and regions collapse into a single point and all Δ_j are zero. But, more urgently, we have to deal with the trivial solutions familiar from multidimensional unfolding (Heiser (1981), De Leeuw (1983), Busing (2010)). For example, we can collapse all category-points into a single point and put all object-points on a sphere with the category-point at its center. Then all distances between category and object-points are the same and we can choose all dissimilarities equal to that distance value and obtain stress zero. It does not help to require $X'X = I$, because we can easily construct orthogonal configurations with all points on the sphere. Stronger normalization constraints are required.

In this paper we use the constraint

$$\sum_{\ell=1}^{k_j}(\delta^j_{i\ell} - \overline{\delta}_{ij})^2 = c_j, \tag{6}$$

with $\overline{\delta}_{ij}$ the mean of the $\delta^j_{i\ell}$ and with c_j constants. For instance we could choose $c_j = 1$ or $c_j = k_j$ for all j. In addition, of course, the $\delta^j_{i\ell}$ must satisfy the ordinal constraints. Constraint (6) explicitly excludes the possibility that the $\delta^j_{i\ell}$ for given i and j are equal for all ℓ.

2 Algorithm

We use an *Alternating Least Squares* or *ALS* algorithm, similar to many other non-metric muldimensional scaling methods, to minimize loss (5). For context, see Borg and Groenen (2005). We alternate minimization over Δ for given X and Y with minimization over X and Y

for given Δ. Using superscript (μ) for the iteration index we can write

$$\Delta^{(\mu)} = \underset{\Delta}{\text{argmin}}\, \sigma(X^{(\mu)}, Y^{(\mu)}, \Delta), \tag{7}$$

$$(X^{(\mu+1)}, Y^{(\mu+1)}) = \underset{X,Y}{\text{argmin}}\, \sigma(X, Y, \Delta^{(\mu)}). \tag{8}$$

The first substep of iteration μ is a form of monotone regression for a very simple partial order, in which one of k_j elements must be smaller than the other elements. The second substep is a metric multidimensional scaling step. It cannot be carried out completely, because it would need an infinite iterative process by itself. Thus the second substep consists of a finite number of *inner iterations* within the second substep of an ALS *outer iteration*. For the inner iterations we use standard smacof majorization steps (De Leeuw (1977a)).

2.1 Monotone Regression

The nm monotone regressions are applied for each object i and each variable j. Forgetting about all these indices for a moment we need to minimize the sum of squares $\|\delta - d\|^2$ over δ, where one of the elements, say δ_ν, must be smaller than or equal to all others, and where $\|\delta - \overline{\delta}\|^2 = c$.

Consider the cone of all vectors δ in $\mathbb{R}^n$ for which $\delta_i \leq \delta_\ell$ for all $i \neq \ell$ and $e'\delta = 0$. The extreme rays of the cone will be the $n-1$ vectors of the form $\alpha e_i + \beta(e - e_i)$ with $i \neq \ell$. Thus element i is equal to α and all other elements (including δ_ℓ) are equal to β. Thus we must have $\alpha > \beta$. The sum of the elements is $\alpha + (n-1)\beta$, which must be zero. Thus $\alpha = -(n-1)\beta$, which implies $\beta < 0$ and $\alpha > 0$. The extreme rays are positive multiples of the vectors $e_i - n^{-1}e$ with $i \neq \ell$ and the cone is the set of all weighted sums with non-negative weights of these $n-1$ vectors. Here is a small example.

```
a <- -cbind(1, -diag(4))
b <- (diag(5)-(1 / 5))[, -1]
print(a)
```

```
     [,1] [,2] [,3] [,4] [,5]
[1,]   -1    1    0    0    0
[2,]   -1    0    1    0    0
[3,]   -1    0    0    1    0
[4,]   -1    0    0    0    1
```

```r
print(b)
```

```
     [,1] [,2] [,3] [,4]
[1,] -0.2 -0.2 -0.2 -0.2
[2,]  0.8 -0.2 -0.2 -0.2
[3,] -0.2  0.8 -0.2 -0.2
[4,] -0.2 -0.2  0.8 -0.2
[5,] -0.2 -0.2 -0.2  0.8
```

```r
print(a %*% b)
```

```
     [,1] [,2] [,3] [,4]
[1,]    1    0    0    0
[2,]    0    1    0    0
[3,]    0    0    1    0
[4,]    0    0    0    1
```

For later reference the sum of squares of the vectors $e_i - n^{-1}e$ is $(n-1)/n$, so the vectors $\sqrt{n/(n-1)}(e_i - n^{-1}e)$ are the vectors of length 1 on the extreme rays.

Suppose y is a vector with n elements. The problem we study is to minimixe the sum of squares $\sigma(x) := \|x-y\|^2$ over $x \in K$ and $x'Jx = 1$. Here

- K is the cone of vectors with $x_1 \leq x_2 \leq \cdots \leq x_n$,
- J is the centering matrix $J := I - n^{-1}ee'$, where
- e is the vector with all n elements equal to one.

Change variables to $\tilde{x} = Jx$. Then $x'Jx = \tilde{x}'\tilde{x}$ and $x \in K$ if and only if $\tilde{x} \in K$. Also $\tilde{x} = Jx$ if and only if $x = \tilde{x} + \beta e$. So the transformed problem is to minimize $\|\tilde{x} + \beta e - y\|^2$ over β and $\tilde{x} \in K$, with $e'\tilde{x} = 0$ and $\tilde{x}'\tilde{x} = 1$. The minimizer for β is $\overline{y} := n^{-1}e'y$, and thus the problem becomes minimizing $\|\tilde{x} - Jy\|^2$ over $\tilde{x} \in K$, $\tilde{x}'\tilde{x} = 1$, and $e'\tilde{x} = 0$.

Forget about the constraint $e'\tilde{x} = 0$. If M is the monotone regression operator it follows that the solution for $\tilde{x}$ is $M(Jy)/\|M(Jy)\|$, which automatically satisfies $e'\tilde{x} = 0$. Thus the solution of our original problem is

$$\hat{x} = \frac{M(y-\overline{y})}{\|M(y-\overline{y})\|} + \overline{y}.$$

$M(y-\overline{y}) = M(y) - \overline{y}$ and thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 + n\overline{y}^2 - 2\overline{y}e'M(y)$ nor $e'M(y) = e'y = n\overline{y}$. Thus $\|M(y-\overline{y})\|^2 = \|M(y)\|^2 - n\overline{M(y)}^2$

$$\hat{x} = \frac{M(y) - \overline{y}e}{\|M(y-\overline{y})\|} + \overline{y}e = \frac{1}{\|M(y-\overline{y})\|}M(y) + \left\{1 - \frac{1}{\|M(y-\overline{y})\|}\right\}\overline{y}e$$

2.2 Majorization

The majorization algorithm for multidimensional scaling

$$A_{ij} := \begin{bmatrix} e_ie_i' & -e_ie_j' \\ -e_je_i' & e_je_j' \end{bmatrix}$$

$$Z := \begin{bmatrix} X \\ Y \end{bmatrix}$$

$$d_{ij}(X,Y) = \text{tr}\, Z'A_{ij}Z$$

$$V := \sum_{i=1}^{n}\sum_{j=1}^{m} A_{ij} = \begin{bmatrix} mI & -ee' \\ -ee' & nI \end{bmatrix} \tag{9}$$

If we define

$$V_1 := (n+m)^{-1}\begin{bmatrix} \frac{m}{n}E & -E \\ -E & \frac{n}{m}E \end{bmatrix}, \tag{10}$$

$$V_2 := \begin{bmatrix} J_n & 0 \\ 0 & 0 \end{bmatrix}, \tag{11}$$

$$V_3 := \begin{bmatrix} 0 & 0 \\ 0 & J_m \end{bmatrix}, \tag{12}$$

then

$$V = (n+m)V_1 + mV_2 + nV_3$$

Matrices V_1, V_2, and V_3 are symmetric projectors on orthogonal subspaces. It follows that V^+, the Moore-Penrose inverse3 of V, is

$$V^+ = \frac{1}{n+m}V_1 + \frac{1}{m}V_2 + \frac{1}{n}V_3$$

$$B(X,Y) := \begin{bmatrix} B_{11}(X,Y) & B_{12}(X,Y) \\ B_{21}(X,Y) & B_{22}(X,Y) \end{bmatrix}$$

2.3 Regression Constraints

2.4 Configuration Constraints

2.4.1 Column-Centering

If we require $X'e = 0$ then the update becomes $J_n\tilde{X}$, with $J_n = I - n^{-1}ee'$ the centering matrix.

This is different from updating X as $J_n\tilde{X}$ and Y as $\tilde{Y} - n^{-1}ee'\tilde{X}$. With this update, in which we subtract the column-means of $\tilde{X}$ from both $\tilde{X}$ and $\tilde{Y}$, the distances are the same before and after centering. Thus the distances in all iterations, and the loss function values, are the same with or without this type of centering.

In general it is advantagous to require $e'X = 0$ because the third term in

$$m\|X - \tilde{X}\|^2 + n\|Y - \tilde{Y}\|^2 - 2\ \mathrm{tr}\ (X - \tilde{X})'ee'(Y - \tilde{Y})$$

becomes independent of X, and thus the (X, Y) minimization is separable.

2.4.2 Orthogonality

If we require $X'X = I$, together with $e'X = 0$, as in Homogeneity Analysis, the solution for X is KL', where $J_n\tilde{X} = K\Lambda L'$ is a singular value decomposition.

2.4.3 Rank Constraints

One of the main themes in Gifi (1990) is that the $k_j \times p$ matrices of category quantifications can be constrained to be of rank $r < p$. We will restrict ourselves in this paper to $r = 1$, which defines what Gifi calls *single quantifications*. If all variables are restricted to have single quantifications then a Homogeneity Analysis becomes equivalent to a Principal Component Analysis of the transformed variables.

In Voronoi Homogeneity Analysis the Voronoi regions of single variables become parallel strips. The category-points of a variable are on a line through the origin, and the Voronoi lines or planes are parallel and perpendicular to the line with category-points.

$$-2\mathrm{tr}\ (X - \tilde{X})'ee'(Y - \tilde{Y}) + n\|Y - \tilde{Y}\|^2 = \mathrm{tr}\ \|(Y - \tilde{Y}) - n^{-1}ee'(X - \tilde{X})\|^2 + C$$

So rank one approximation of $\tilde{Y} + ee'(X - \tilde{X})$. If X is column-centered then of course $\tilde{Y} - ee'\tilde{X})$.

2.4.4 Centroid

A more complicated restriction is $Y=D^{-1}G'X.$ This eliminates Y from the problem and only requires minimization of $

$$m\|X-\tilde X\|^2+n\|D^{-1}G'X-\tilde Y\|^2-2\text{tr }(X-\tilde X)'ee'(D^{-1}G'X-\tilde Y)$$

$$m\text{ tr }X'X+n\text{ tr }X'GD^{-2}G'X-2\text{ tr }X'e_ne_m'D^{-1}G'X.$$

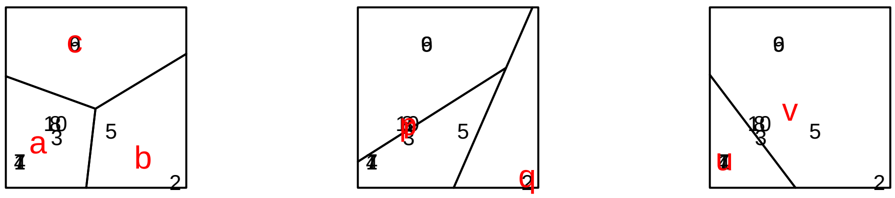
$$-2m\text{ tr }X'\tilde X-2n\text{ tr }X'GD^{-1}\tilde Y+2\text{tr }X'ee'\tilde Y+2\text{tr }X'GD^{-1}ee'\tilde X$$

3 Software

4 Examples

4.1 Small Example

We start with the small example we used before to illustrate Homogeneity Analysis.



Loss is 0.0000000000 after 308 iterations. If we compare the plots from the Voronoi analysis with the star plots from the initial Homogeneity Analysis for this example we see that the algorithm did not have to change much to obtain perfect fit. One thing that does change is that the iterations moves the category-points “p” and “r” for variable 2 very close to each other. At the point where we stop the iterations they are still distinguishable, and thus their perpendicular bisector is still a line in the Voronoi plot. If we would continue iterating “p” and “r” would coincide and there would only be two Voronoi regions: one containing category-point “q” and object-point 2, and one containing all other points. one containing all other

4.2 GALO

Loss is 0.0008972348 after 1000 iterations.

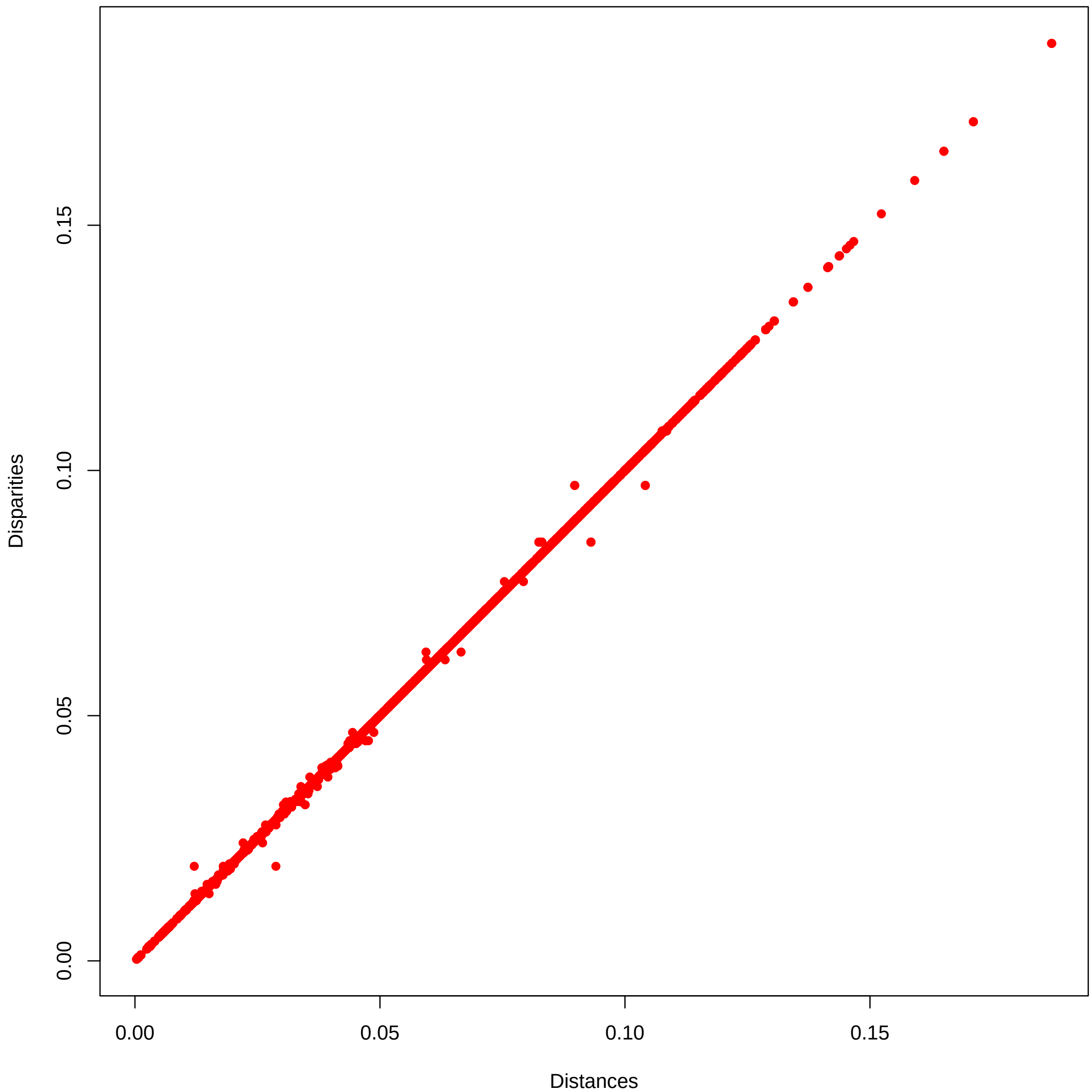


Figure 1: Galo D-Dhat Plot

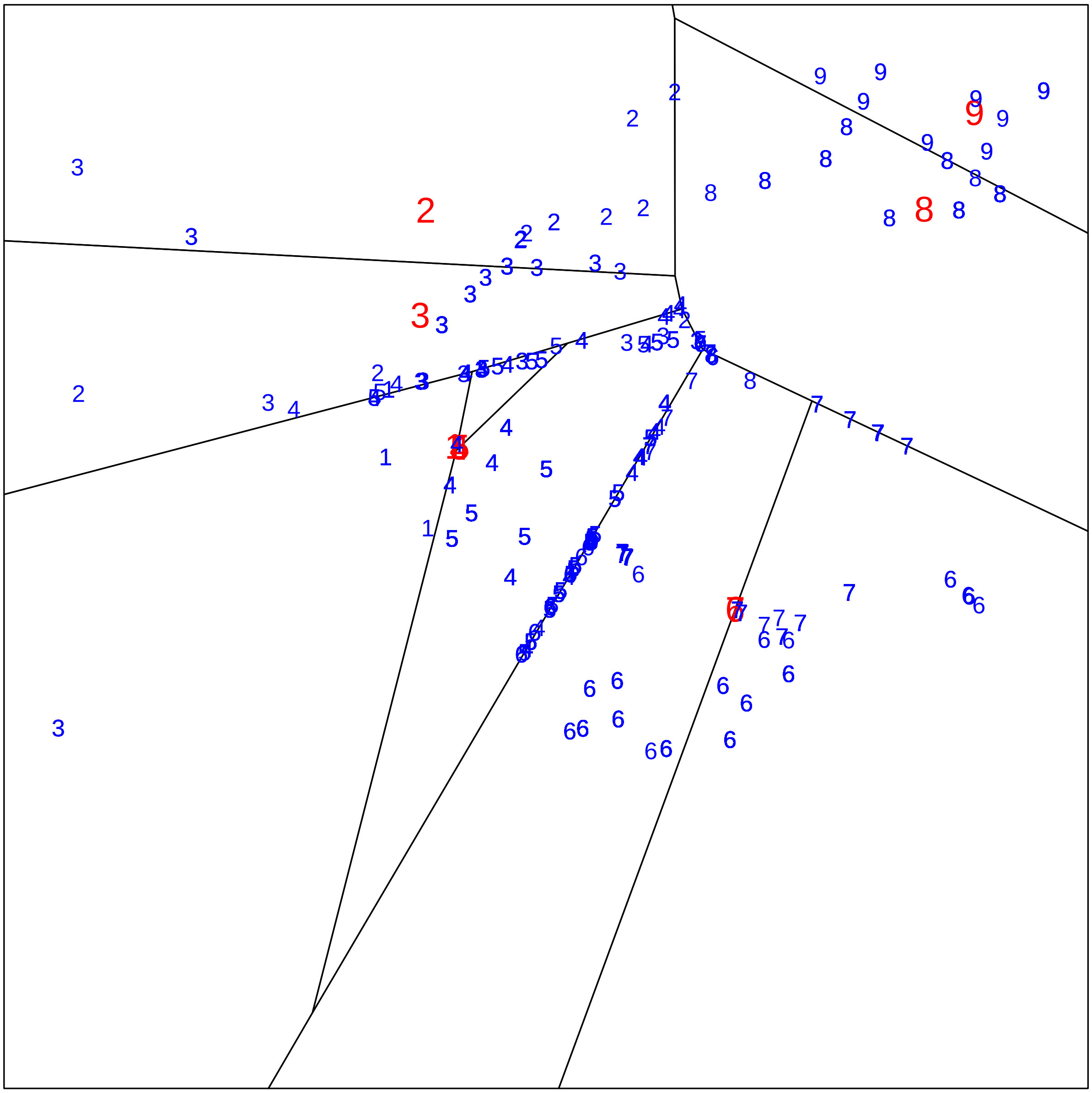


Figure 2: Galo IQ

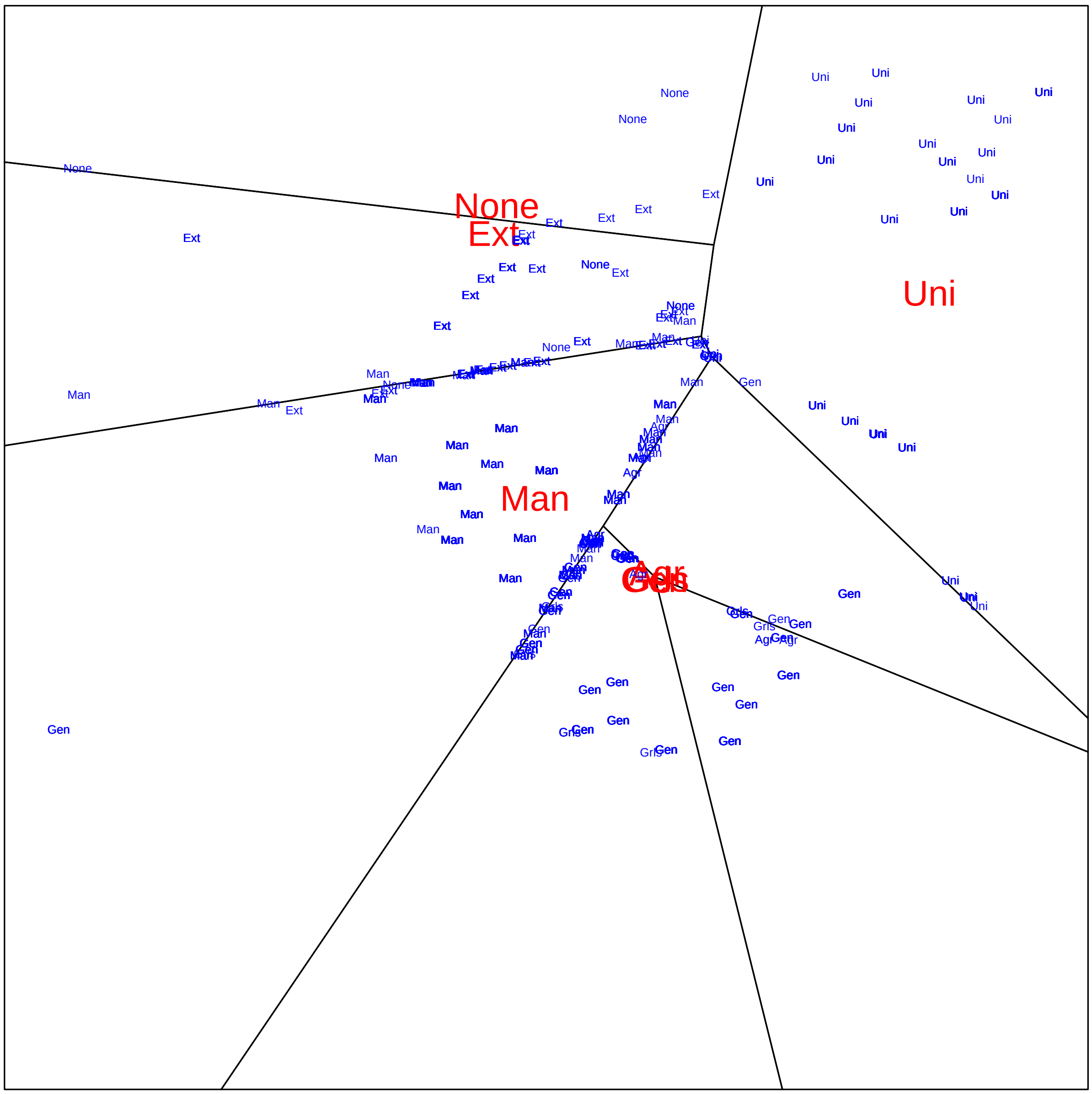


Figure 3: Galo ADV

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